SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Answer Script Evaluation Sheet

Continuous Internal Assessment 1/2

Date:

Programme	B.E/ B. Tech	**Slot**	C / D
Course Code	CSA0709 / CSA0710	**Course Name**	Computer Networks
Duration	1.30 Hrs	**Marks**	50

Q. No.	Max. Marks	Awarded Marks
1.	10	8
2.	10	8
3.	10	8
4.	10	8
5.	10	8
Total	50	40

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1,

(a) Data link layer:

The data link layer is responsible for a transferring data between two directly connected devices over a network

Functions:

1, Framing: Divides data into frames for transmission

2, Physical addressing: uses MAC address to identify devices on local network

3, Error detection: Dectes transmission error using methods like CRC.

4, Flow control: Helps prevent a sender from overhelming a reciver when supported by protocol

2. Transport layer:

The transport layer provides communication between application running on different devices

*, Segmentation: Divider application data into Smaller segments

*, Error recovery: Tcp dectes missing data

Feature	Connection - oriented Service	Connection less Service
Protocol	TCP	UDP
Connected Setup	Required	Not required
Reliability	Support reliable delivery	No delivery guarantee
Acknowledgement	uses acknowledgement	No built in
Data ordering	Maintainare ordered delivery	No ordering guarantee
Speed and overhead	Generally more overhead	Generally lower head
Example	File transfer web browsing using HTTS over TCP	Online gaming dive Steaming using UDP

TCP is preferred when reliable, ordered delivery is required, whereas UDP is useful when low overhead are important.

2,

a) **Encapsulation**:

It is a process of adding protocol information to data as it moves down the OSI layers at the source devices

OSI Layer	Process	Data unit
7. Application	Creates application data	Data
6. Presentation	Handles translation encryption or the compression as needed	Data
5. Session	Manages communications sessions	Data
4. transport	Adds transport header, such as TCP or UDP	Segment
3. Network	Adds MAC addressing and frame checking	Packet
2. Data link	Address source	frame
1. Physical	Transmits bits over the medium	Bits

3, classful IP Addressing:

classfull addressing, IPV4 address were divided into fixed class A, B, C and D and E.

Class	First octet Range	Default Mask	Purpose
A	1 - 126	255.00 (18)	Large network
B	128 - 191	255.255.00 (116)	Medium Sized network
C	192 - 223	255.255.0 (184)	Small network
D	224 - 239	Not applicable	Multicast
E	240 - 255	Not applicable	Reversed

CIDR improves IPV4 address utilization and routing efficientay by using flexible network Prefix lenght Smaller organization can receive Smaller address block.

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Given,

* Network address : 172.16.0.0./16
* Required Subnet : Atleast 6

Step 1 :

$2^n \geq$ Required Subnet

for 6 Subnets :

$2^2 = 4$

$2^3 = 8$

$\therefore$ 3 bits must be borrowed

Step 2 :

Prefix = /16

Borowed = 3

New prefix = 16 + 3 = 19

$\therefore$ CIDR notation : /19

Submasked : 255 : 255 : 224 : 0

5,

Static routing relies on fixed, manually configured paths, dynamic routing automatically calculates route using protocol like RIP, OSPF and EIGRP based on current network topologies condition.

Parameter	Static Routing	Dynamic Routing (RIP, OSPF, EIGRP)
Effort	Manual Setup by admin	Automatic neighbor discovery.
Adaptability	None; requires manual fix if link fails	High: Self-heal around link failure
Convergence	N/A (Static lines donot adapt)	Variables (Seconds to minute)
Scalability	Low, Poor choice	High: Scales to Enterprise
Resource	Minimal CPU and memory use	Higher CPU/Ram use
overhead	Zero routing update traffic	Consumes bandwith for protocol packets
Security	Higher: no rouge route injection	Lower: requires explicit authentication
AD default	1 (highest protocol Preference)	Varies: (EIGRP: 90, OSPF: 110, RIP: 120)